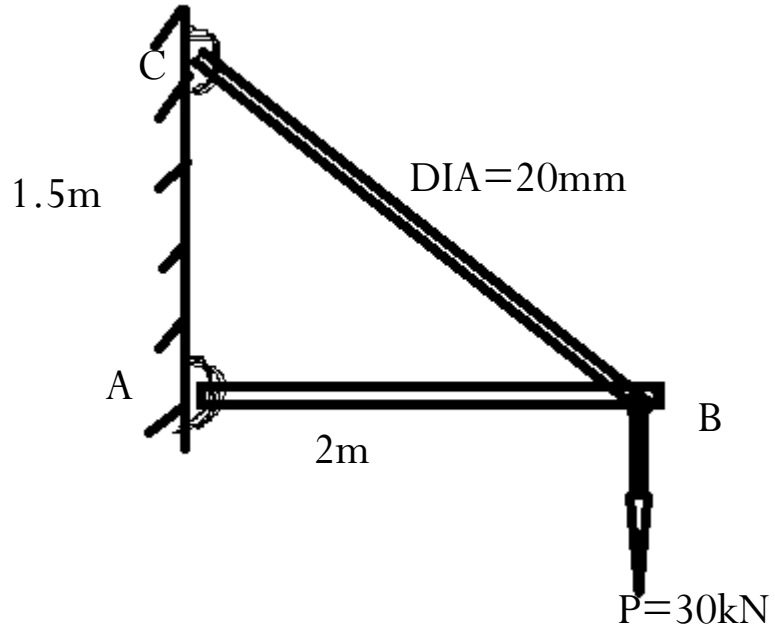
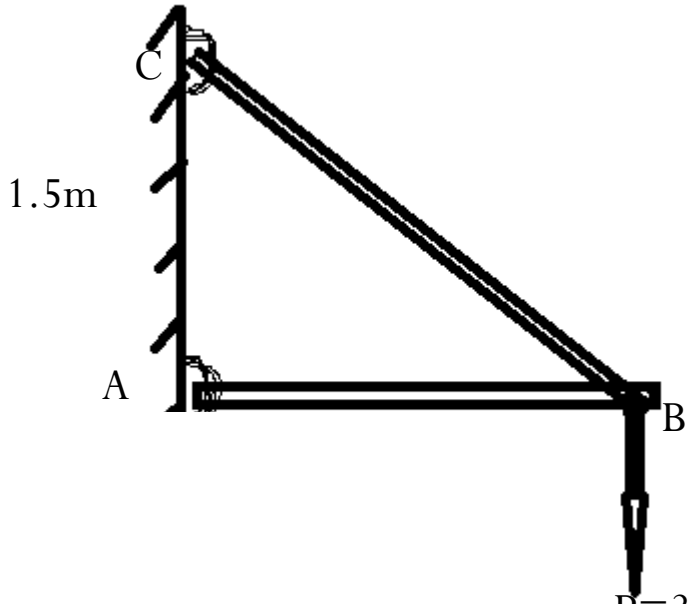
Stress and strain

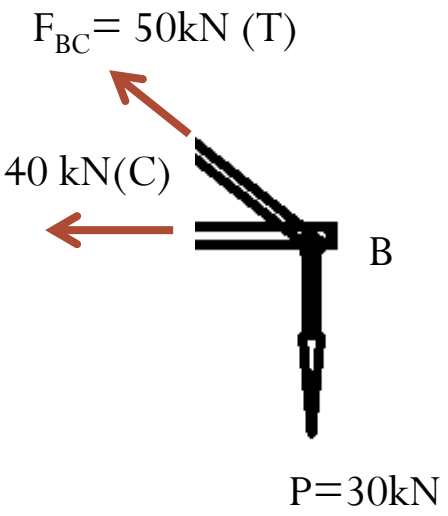
Chapter 4

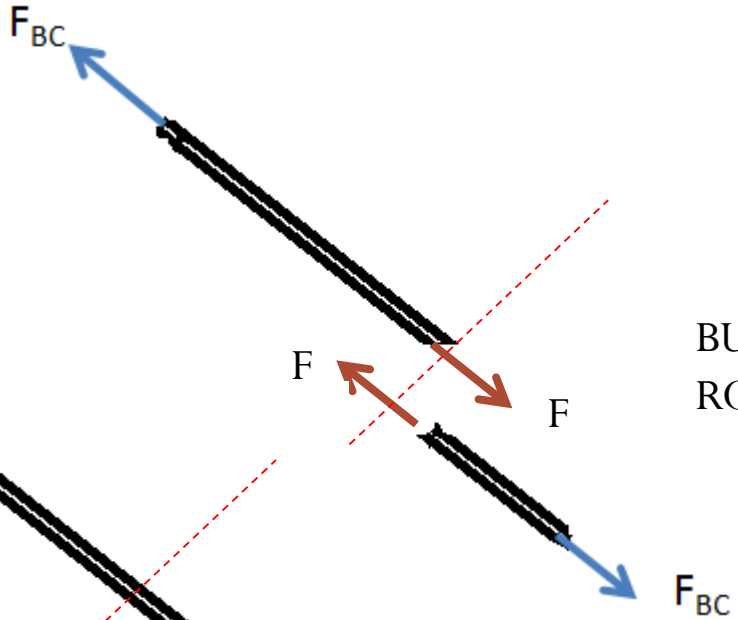


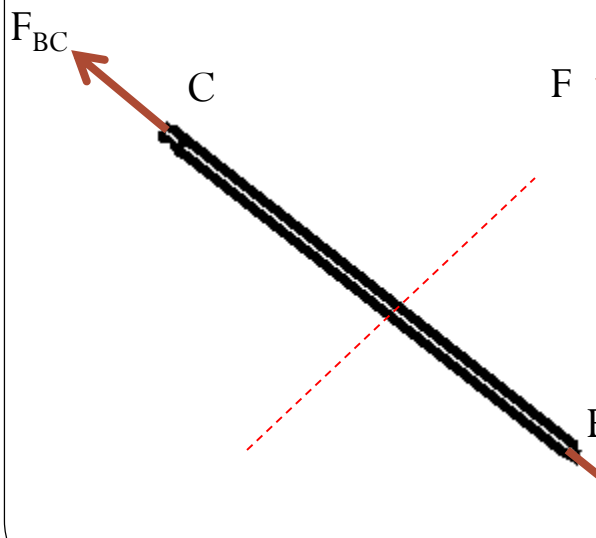
Analysis and design of this structure involve the determination of stresses and deformation

To determine whether a load of 30kN may safely be supported at B by this structure









BUT STILL IT CANNOT BE SAID WHETHER ROD BC CAN SUPPORT THIS LOAD

IT DEPENDS UPON:

1. INTERNAL FORCES
2. CROSS-SECTIONAL AREA OF ROD BC
3. MATERIAL OF ROD

STRESS UNDER GENERAL LOADING

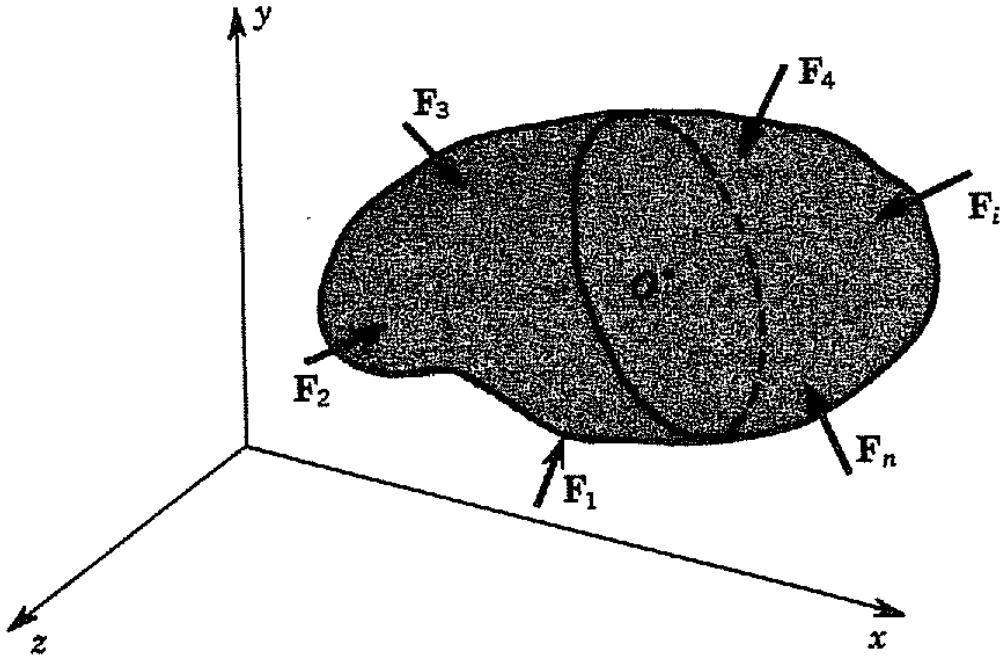


Fig. 4.1 Continuous body acted on by external forces.

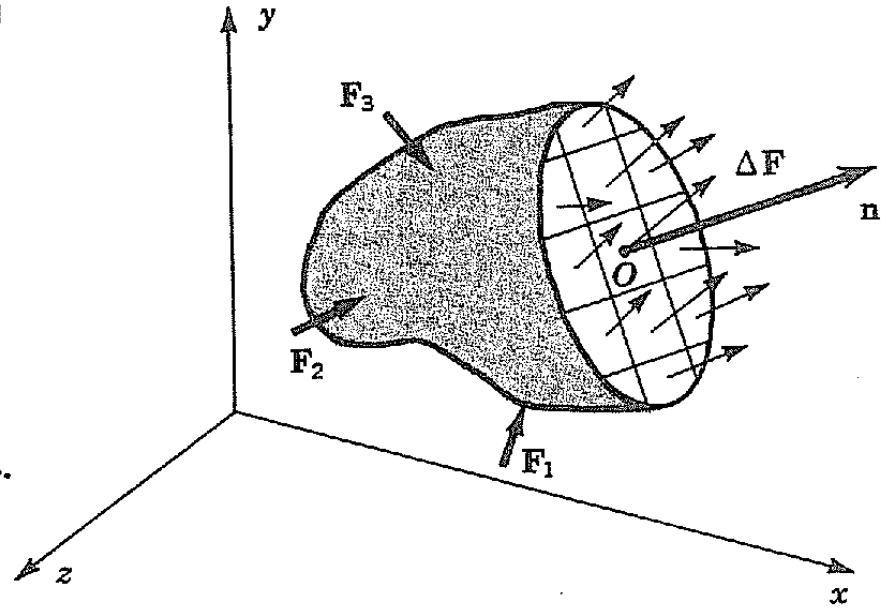


Fig. 4.2 Internal forces acting on a plane whose normal is **n**.

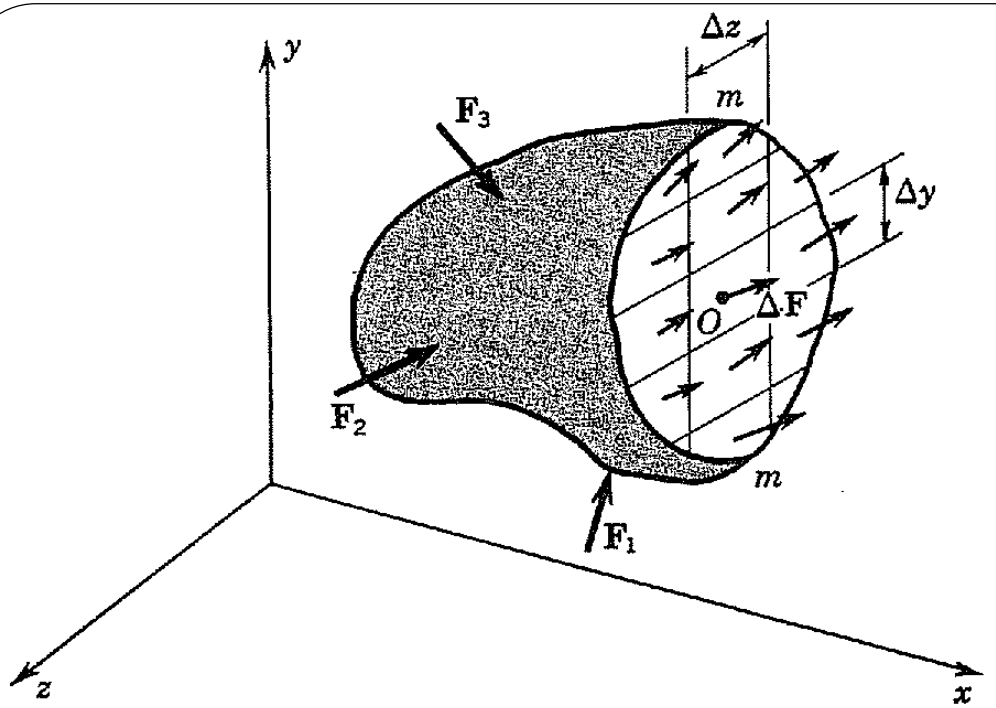


Fig. 4.4 Internal forces acting on plane *mm*.

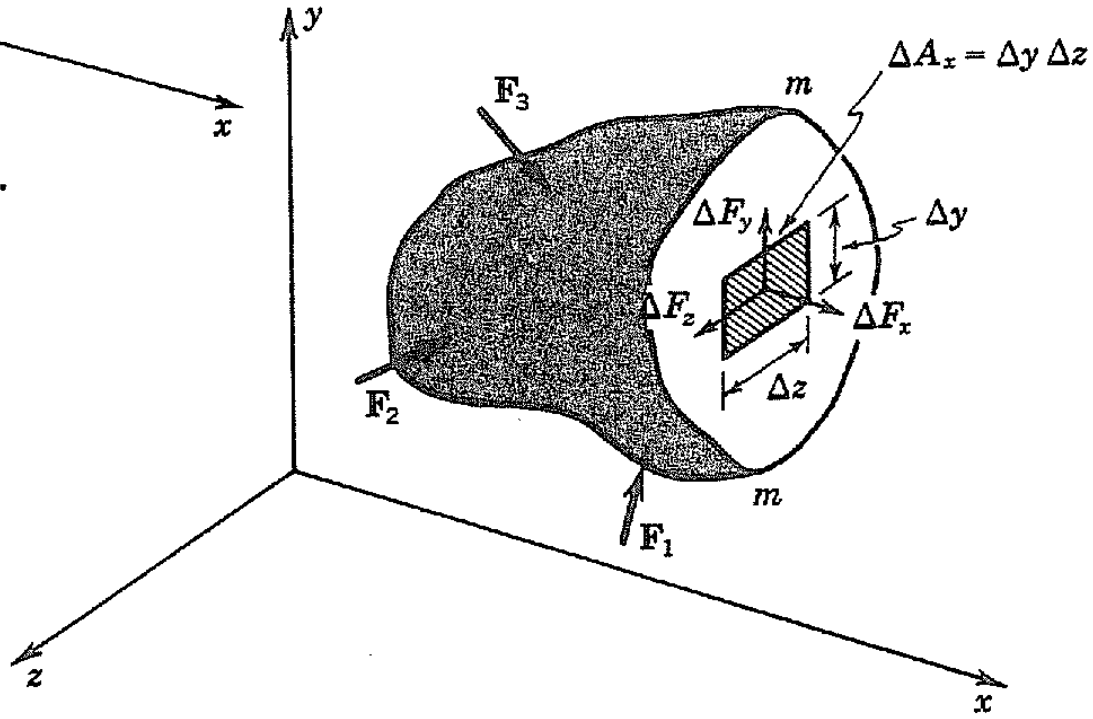
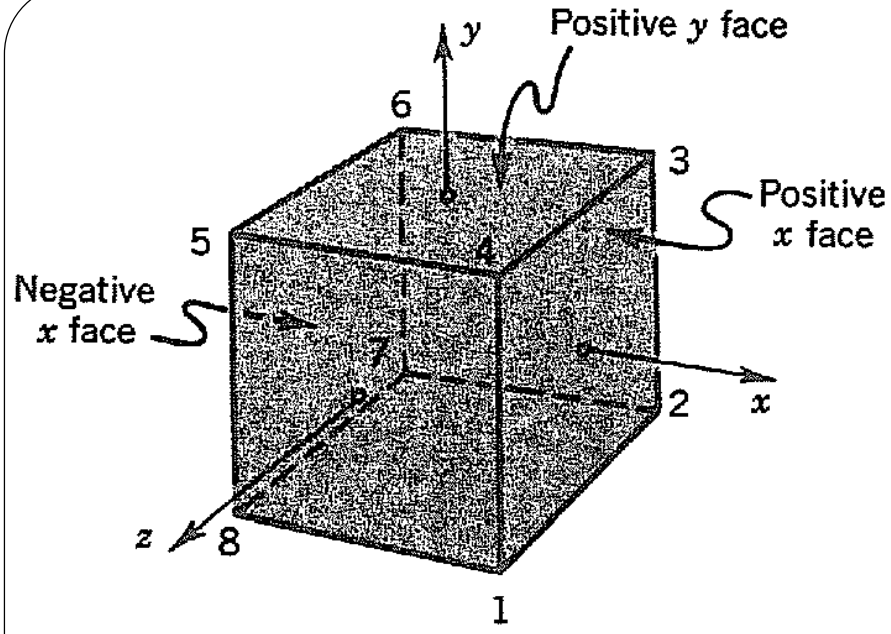


Fig. 4.5 Rectangular components of the force vector ΔF acting on the small area centered on point *O*.



$$\sigma_x = \lim_{\Delta A_x \to 0} \frac{\Delta F_x}{\Delta A_x}, \quad \tau_{xy} = \lim_{\Delta A_x \to 0} \frac{\Delta F_y}{\Delta A_x}, \quad \tau_{xz} = \lim_{\Delta A_x \to 0} \frac{\Delta F_z}{\Delta A_x}$$

$$\begin{matrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{matrix}.$$

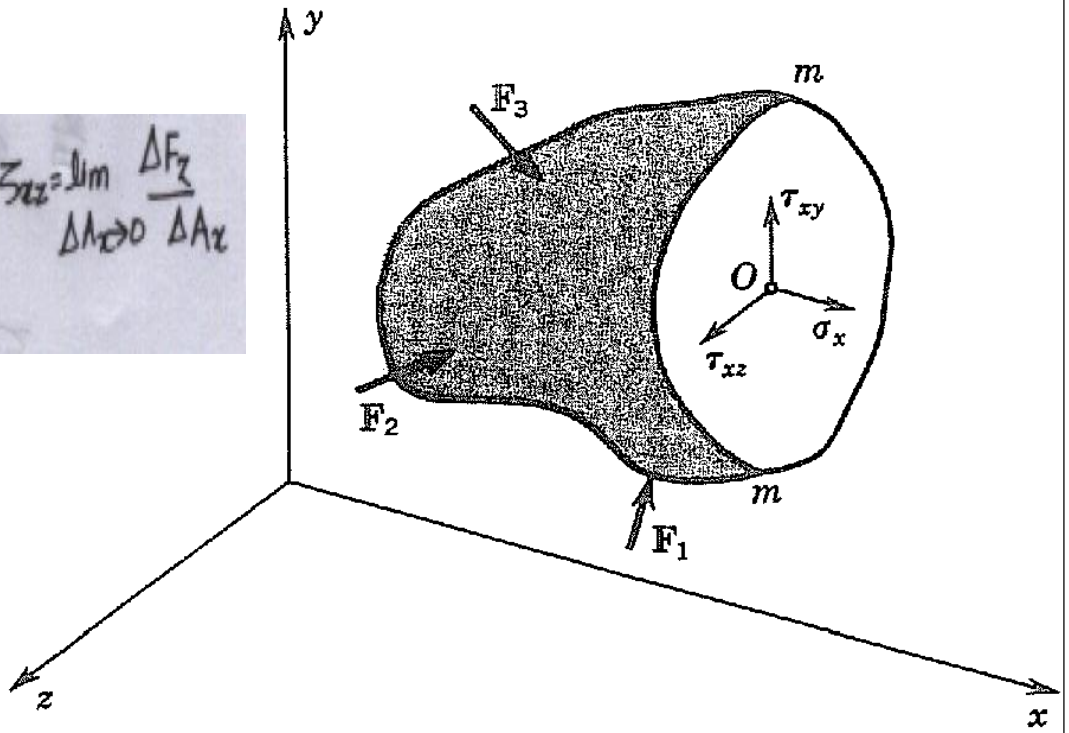


Fig. 4.7 Stress components on positive x face at point O.

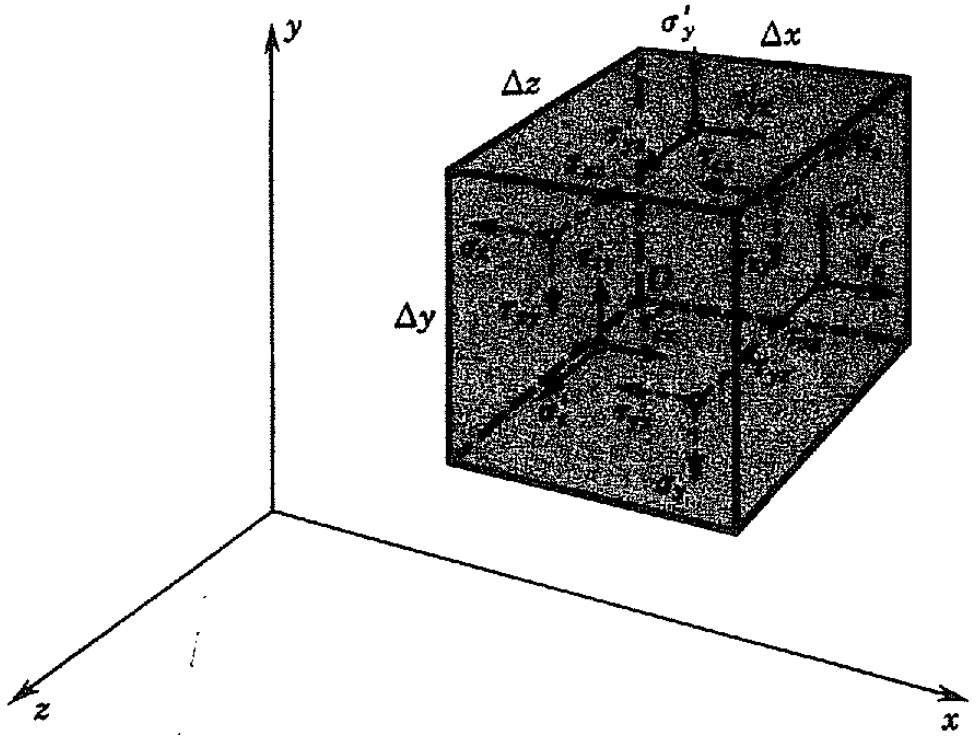


Fig. 4.8 Stress components acting on the six sides of a parallelepiped.

PLANE STRESS

If the stress state at a material particle is such that the only non-zero stress components act in one plane only, the particle is said to be in plane stress.

$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

$$\begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix}$$

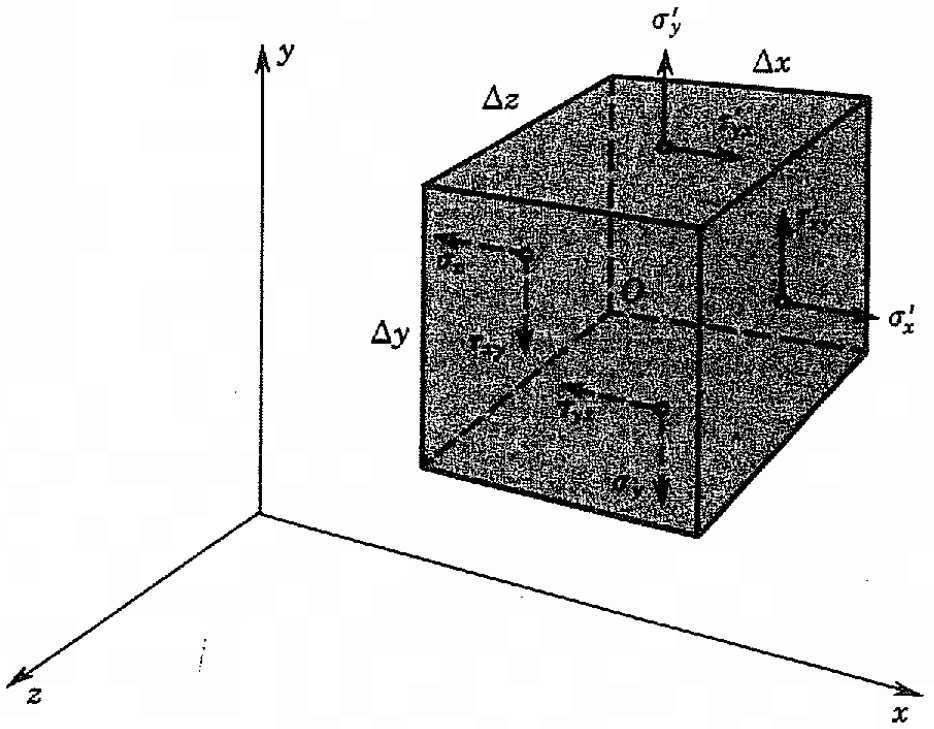
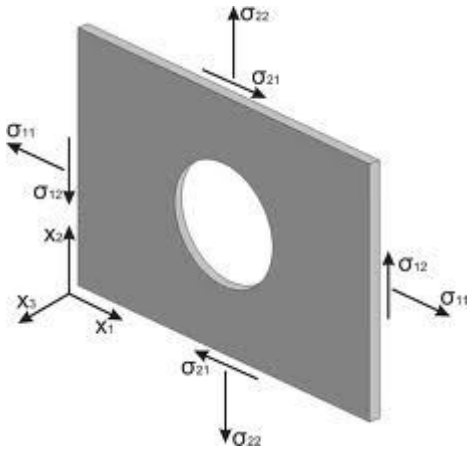
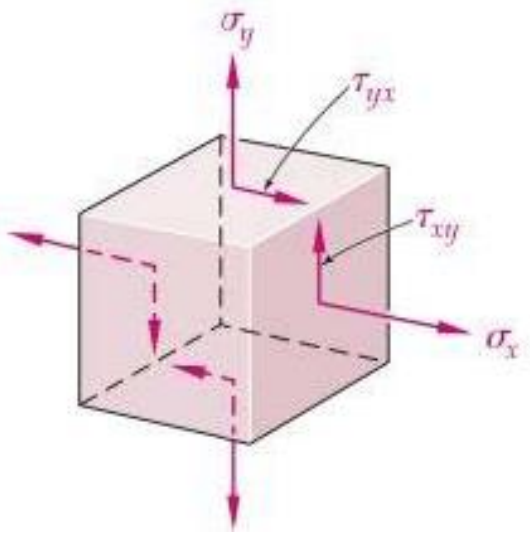
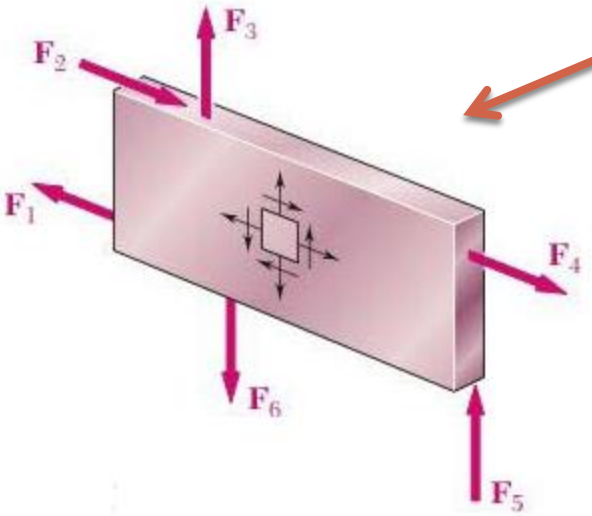


Fig. 4.9 Stress components which define a condition of plane stress in the *xy* plane.





Plane stress i.e. a situation in which two of the faces of the cubic element are free of any stress



Equilibrium of a differential element in plane stress

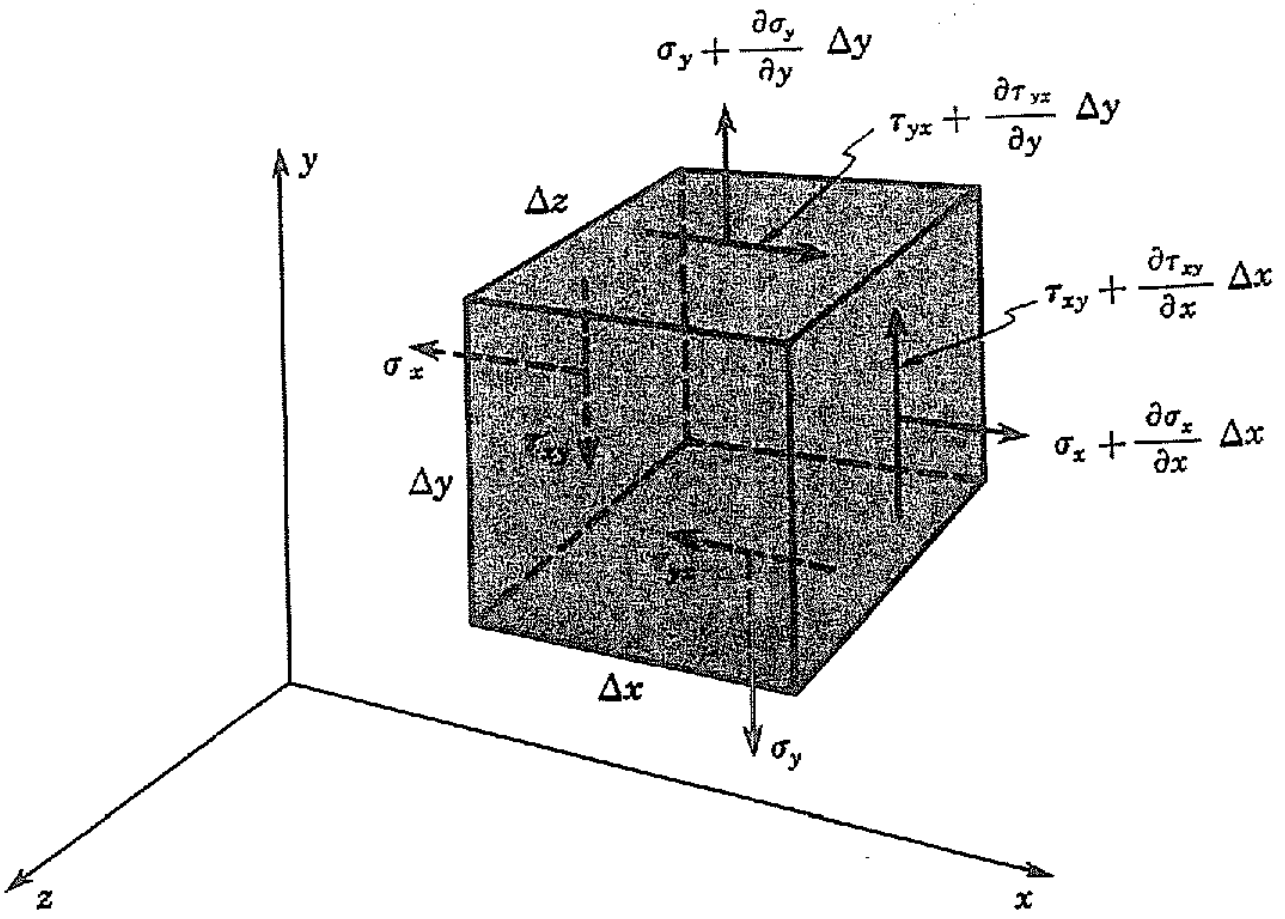


Fig. 4.10 Stress components in plane stress expressed in terms of partial derivatives.

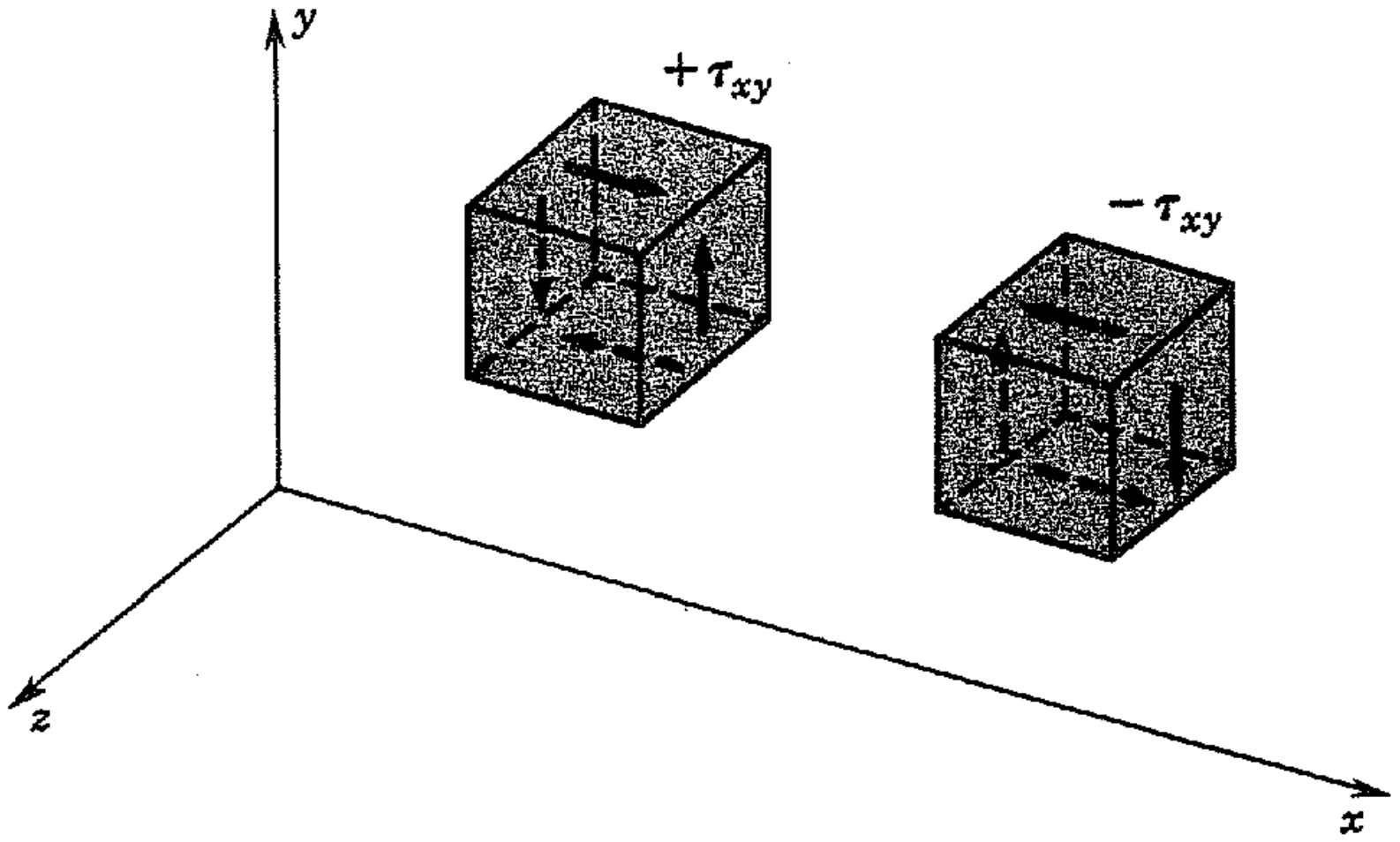


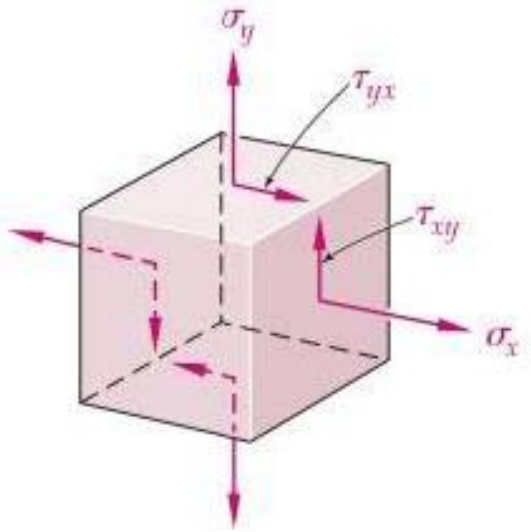
Fig. 4.11 Definition of positive and negative τ_{xy}.

Equilibrium of a differential element in plane stress

$$\tau_{xy} = \tau_{yx}$$

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0$$

$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} = 0$$



Stress components associated with arbitrarily oriented faces in plane stress

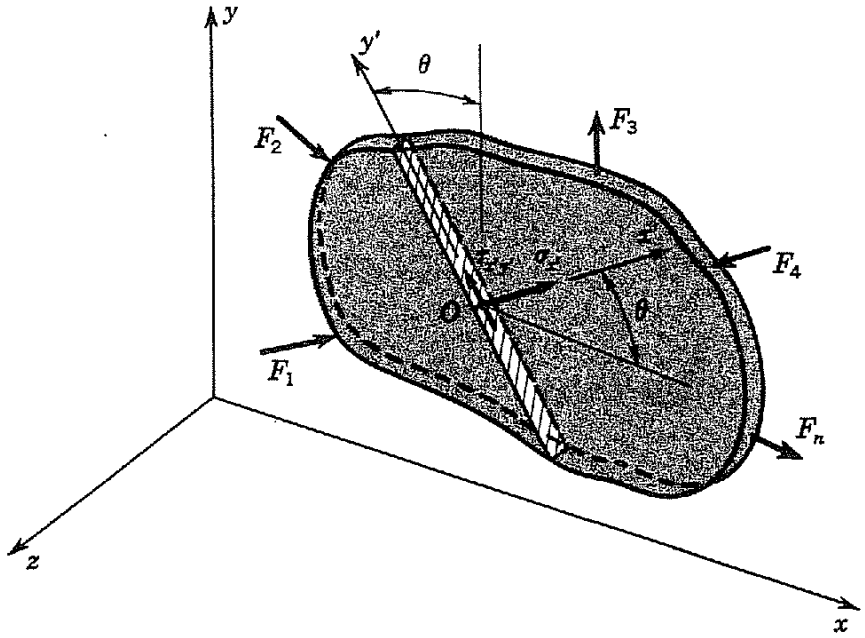


Fig. 4.14 Thin body subjected to plane stress in the xy plane. x' and y' are a set of perpendicular axes lying in the plane of the body, and $\sigma_{x'}$, $\tau_{x'y'}$ are the stress components acting on the positive x' face at point O.

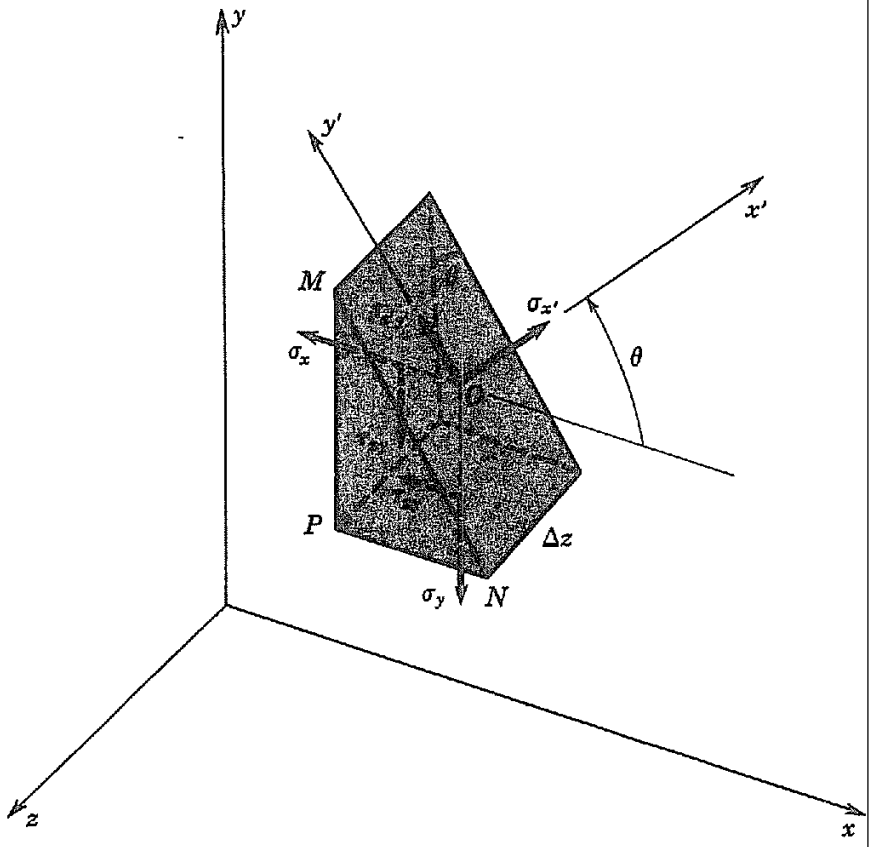
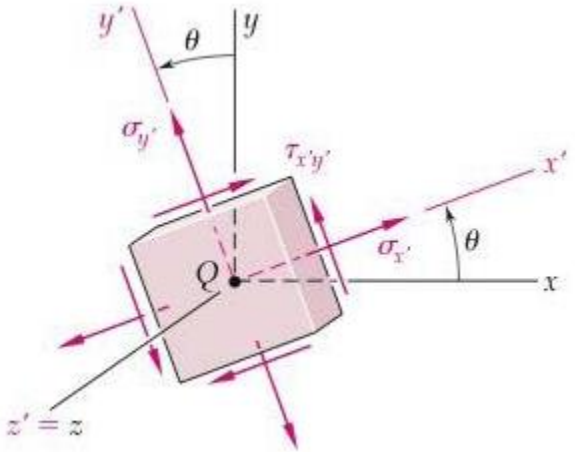
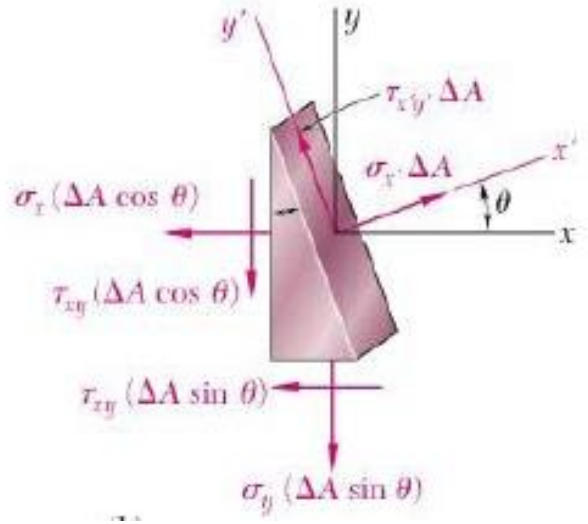


Fig. 4.15 Stress components acting on faces of a small wedge, cut from body of Fig. 4.14, which is in a state of plane stress in the xy plane.

Stress components associated with arbitrarily oriented faces in plane stress





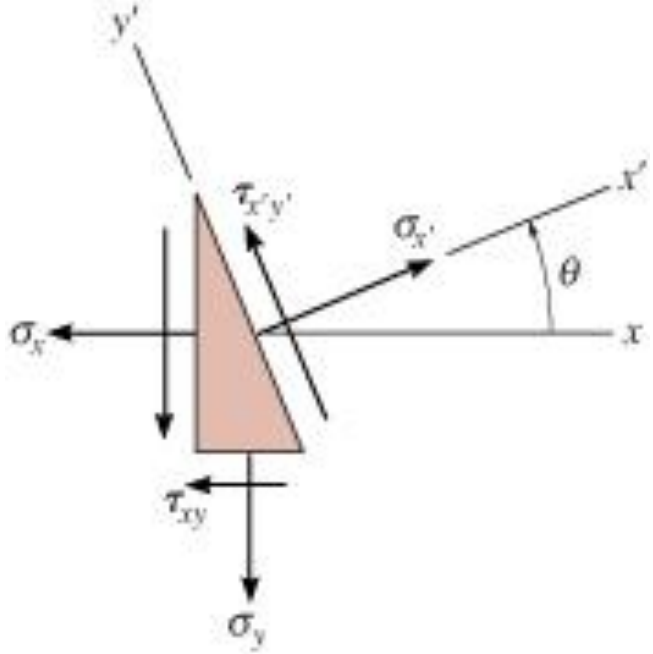
$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

MOHR'S CIRCLE: PLANE STRESS

- Equations for plane stress transformation have a graphical solution that is easy to remember and use.
- This approach will help you to "visualize" how the normal and shear stress components vary as the plane acted on is oriented in different directions.



(a)

MOHR'S CIRCLE: PLANE STRESS

- Eqns are written as

$$\sigma_{x'} = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

- Parameter can be eliminated by squaring each eqn and adding them together.

$$\left[\sigma_{x'} - \left(\frac{\sigma_x + \sigma_y}{2}\right)\right]^2 + \tau^2{}_{x'y'} = \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2{}_{xy}$$

MOHR'S CIRCLE: PLANE STRESS

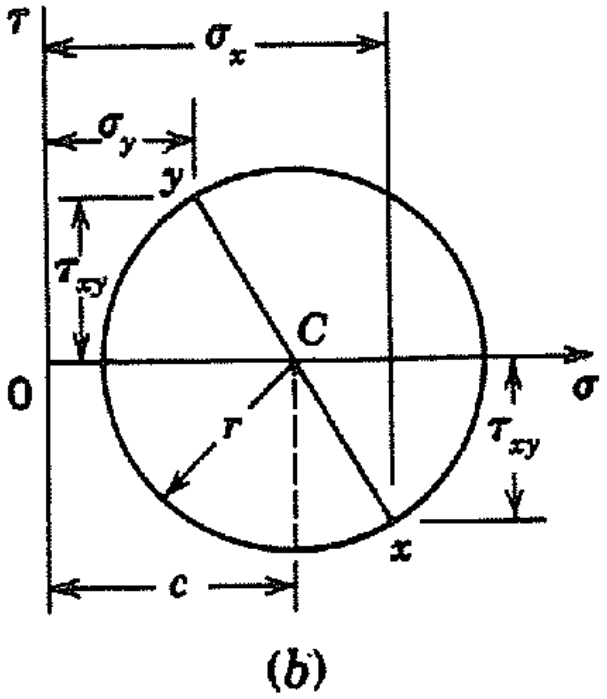
- If σ_x, σ_y, τ_{xy} are known constants, thus we compact the Eqn as,

$$\left(\sigma_{x'} - \sigma_{avg}\right)^2 + \tau^2_{x'y'} = R^2$$

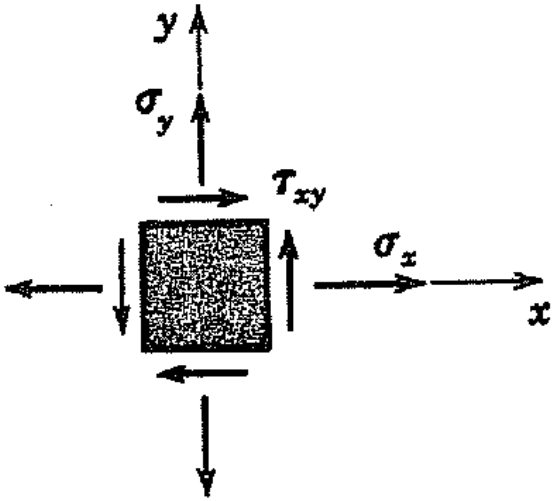
where

$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2_{xy}}$$

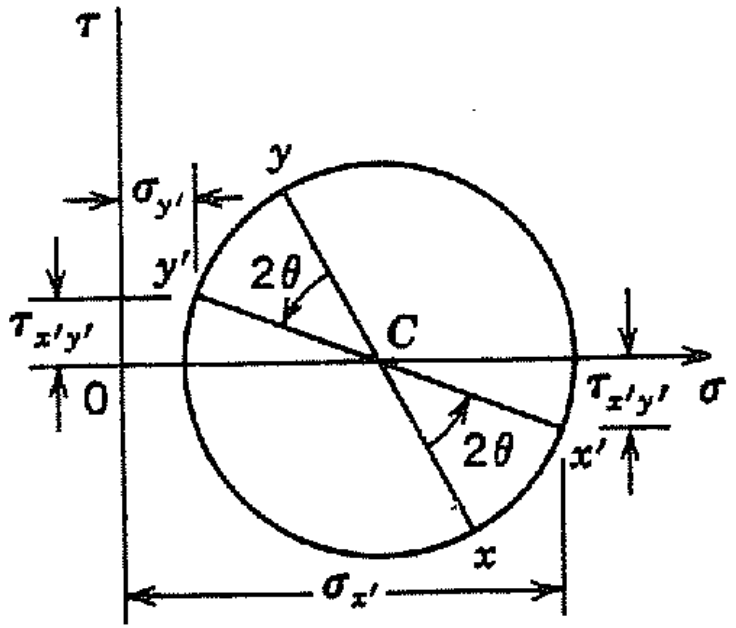


(b)

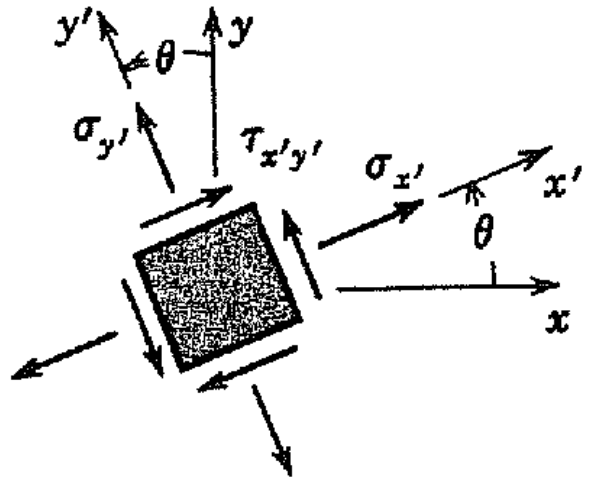


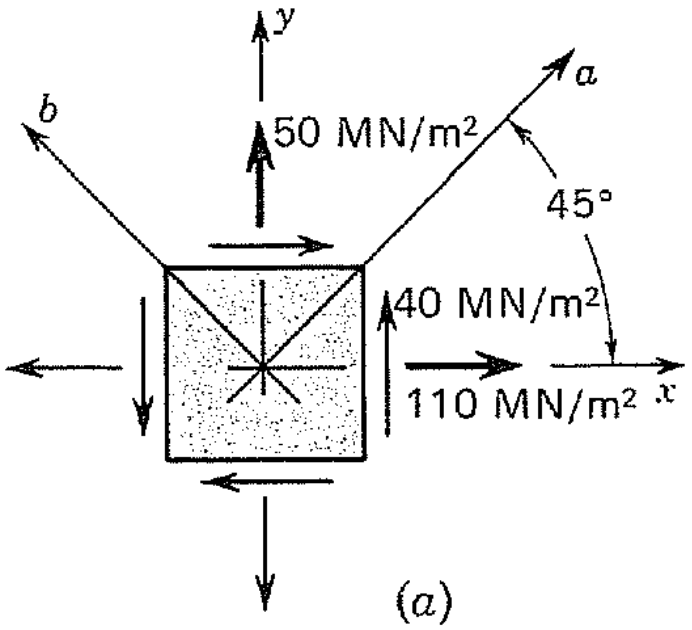
- For normal stress:
  - Tension is positive
    - Plotted right to the origin of the stress plane
  - Compression is negative
    - Plotted left to the origin of the stress plane

- For shear stress:
  - Positive shear stress is plotted downward for x face and upward for y face
  - Negative shear stress is plotted upward for x face and downward for y face



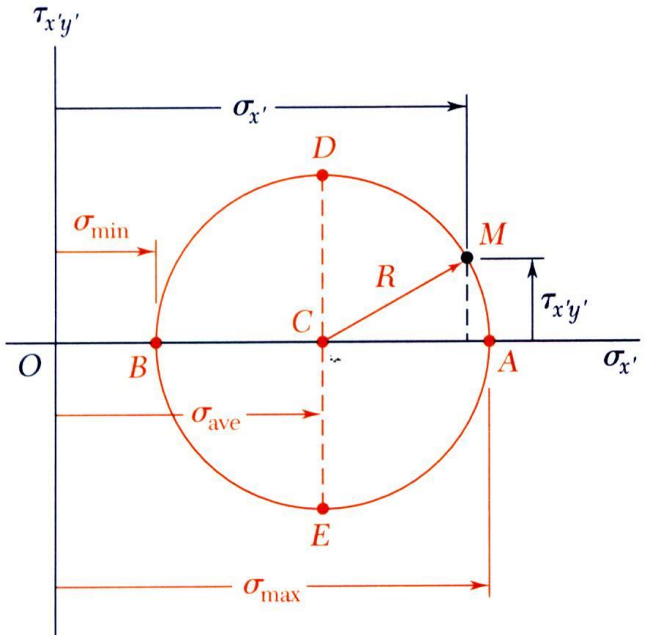
(c)

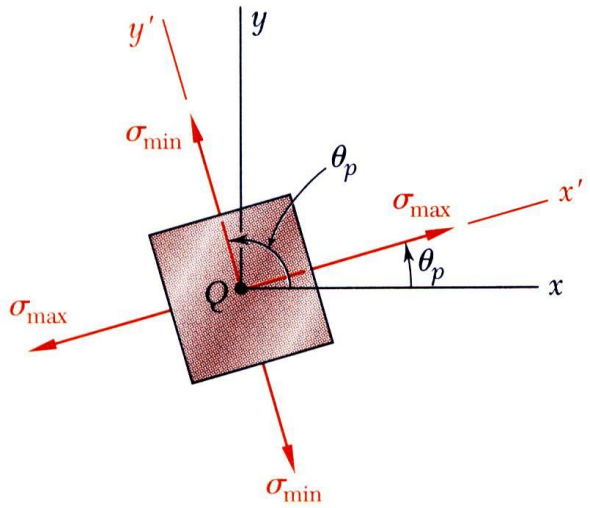




- Find the stress components of a plane whose normal is inclined at 45° to the xy axes.

Principal Stresses and Principal Planes





- The previous equations are combined to yield parametric equations for a circle,

$$(\sigma_{x'} - \sigma_{ave})^2 + \tau_{x'y'}^2 = R^2$$

where

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} \qquad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

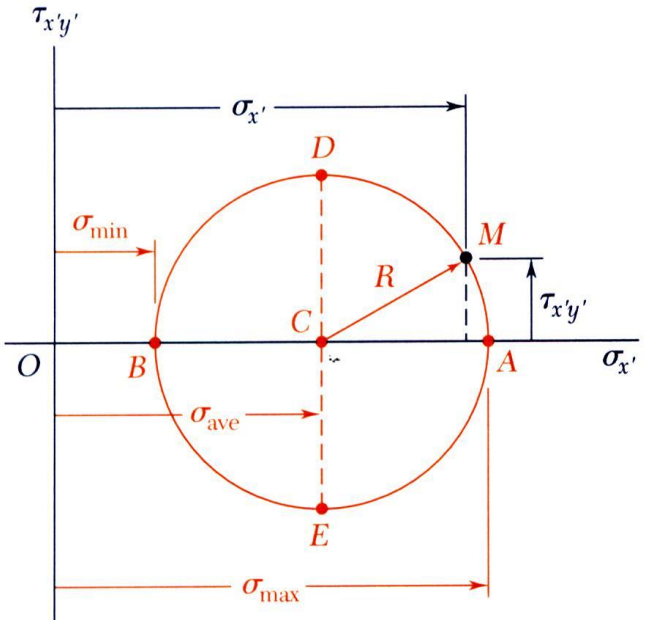
- *Principal stresses* occur on the *principal planes of stress* with zero shearing stresses.

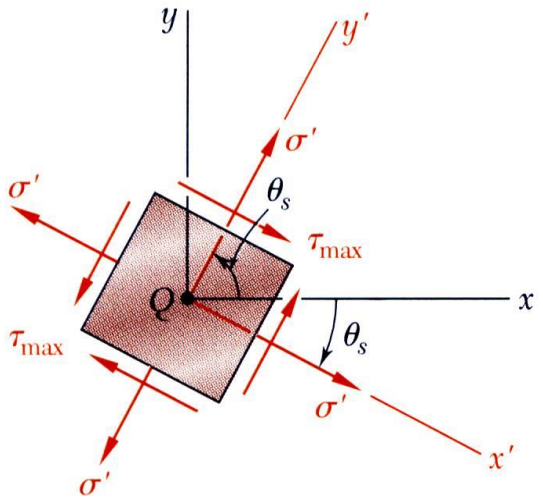
$$\sigma_{max,\,min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

Note : defines two angles separated by 90°

Maximum Shearing Stress





Maximum shearing stress occurs for $\sigma_{x'} = \sigma_{ave}$

$$\tau_{\max} = R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}}$$

Note : defines two angles separated by 90° and offset from θ_p by 45°

$$\sigma' = \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

Example

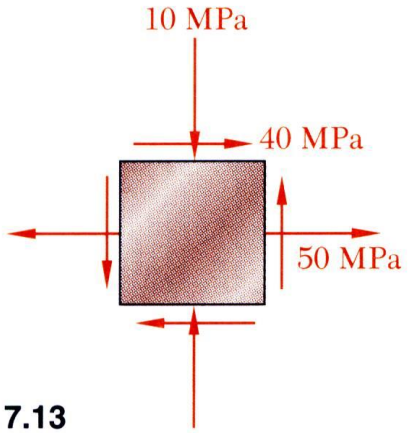


Fig. 7.13

For the state of plane stress shown, determine (a) the principal planes, (b) the principal stresses, (c) the maximum shearing stress and the corresponding normal stress.

SOLUTION:

- Find the element orientation for the principal stresses from

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

- Determine the principal stresses from

$$\sigma_{\max,\min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- Calculate the maximum shearing stress with

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma' = \frac{\sigma_x + \sigma_y}{2}$$

Example

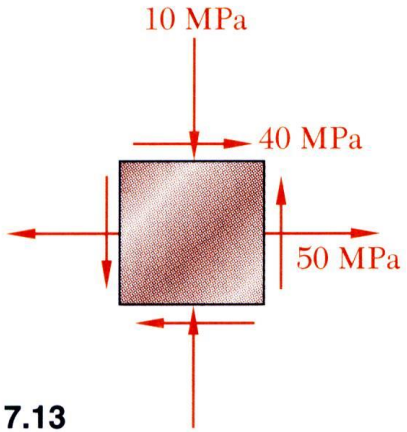


Fig. 7.13

$\sigma_x = +50\,\text{MPa}$ $\tau_{xy} = +40\,\text{MPa}$

$\sigma_y = -10\,\text{MPa}$

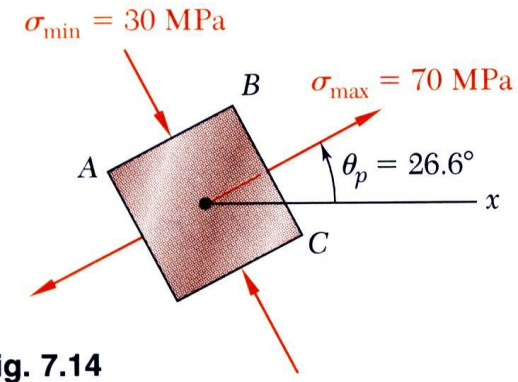


Fig. 7.14

SOLUTION:

- Find the element orientation for the principal stresses from

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2(+40)}{50-(-10)} = 1.333$$

$$2\theta_p = 53.1°,\ 233.1°$$

$$\boxed{\theta_p = 26.6°,\ 116.6°}$$

- Determine the principal stresses from

$$\sigma_{\max,\min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 20 \pm \sqrt{(30)^2 + (40)^2}$$

$$\boxed{\begin{aligned}\sigma_{\max} &= 70\,\text{MPa}\\ \sigma_{\min} &= -30\,\text{MPa}\end{aligned}}$$

Example

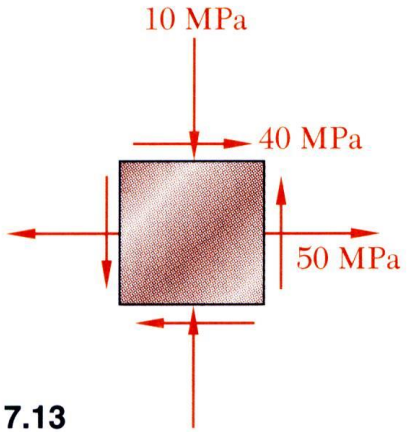


Fig. 7.13

$\sigma_x = +50\,\text{MPa}$ $\tau_{xy} = +40\,\text{MPa}$

$\sigma y = -10\,\text{MPa}$

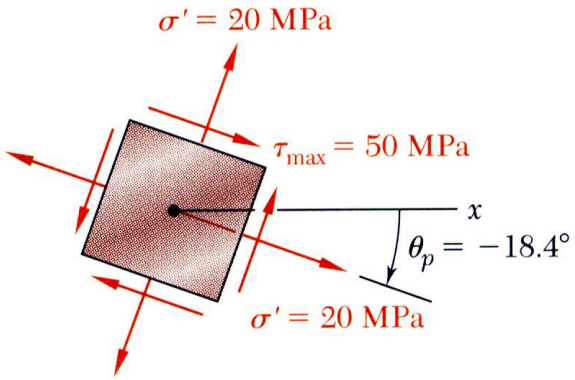


Fig. 7.16

- Calculate the maximum shearing stress with

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$\boxed{\tau_{\max} = 50\,\text{MPa}}$$

$$\theta_s = \theta_p - 45$$

$$\boxed{\theta_s = -18.4°,\ 71.6°}$$

- The corresponding normal stress is

$$\sigma' = \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{50 - 10}{2}$$

$$\boxed{\sigma' = 20\,\text{MPa}}$$

Example

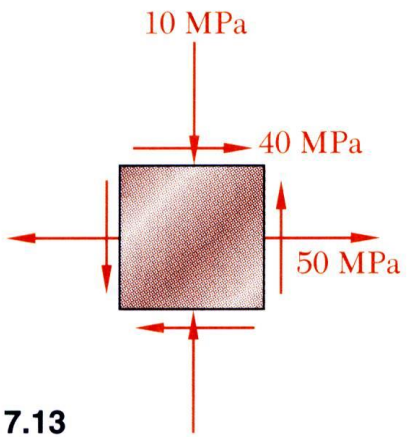
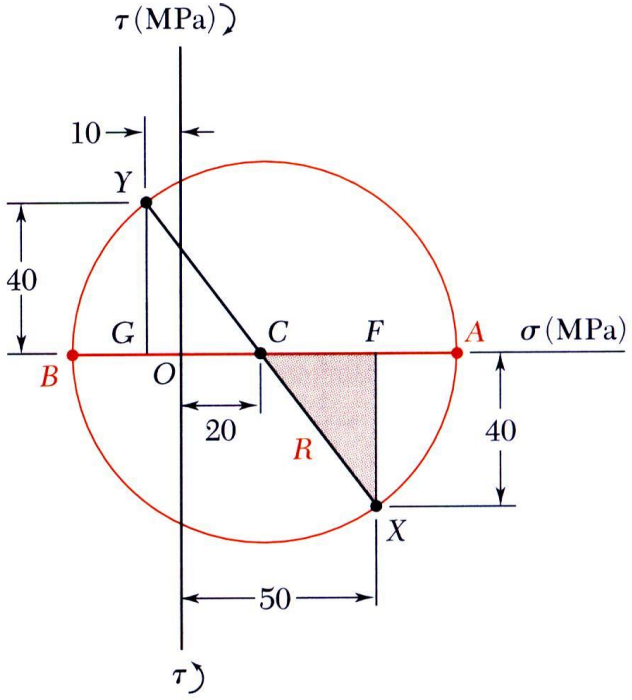


Fig. 7.13

For the state of plane stress shown, (a) construct Mohr's circle, determine (b) the principal planes, (c) the principal stresses, (d) the maximum shearing stress and the corresponding normal stress.



SOLUTION:

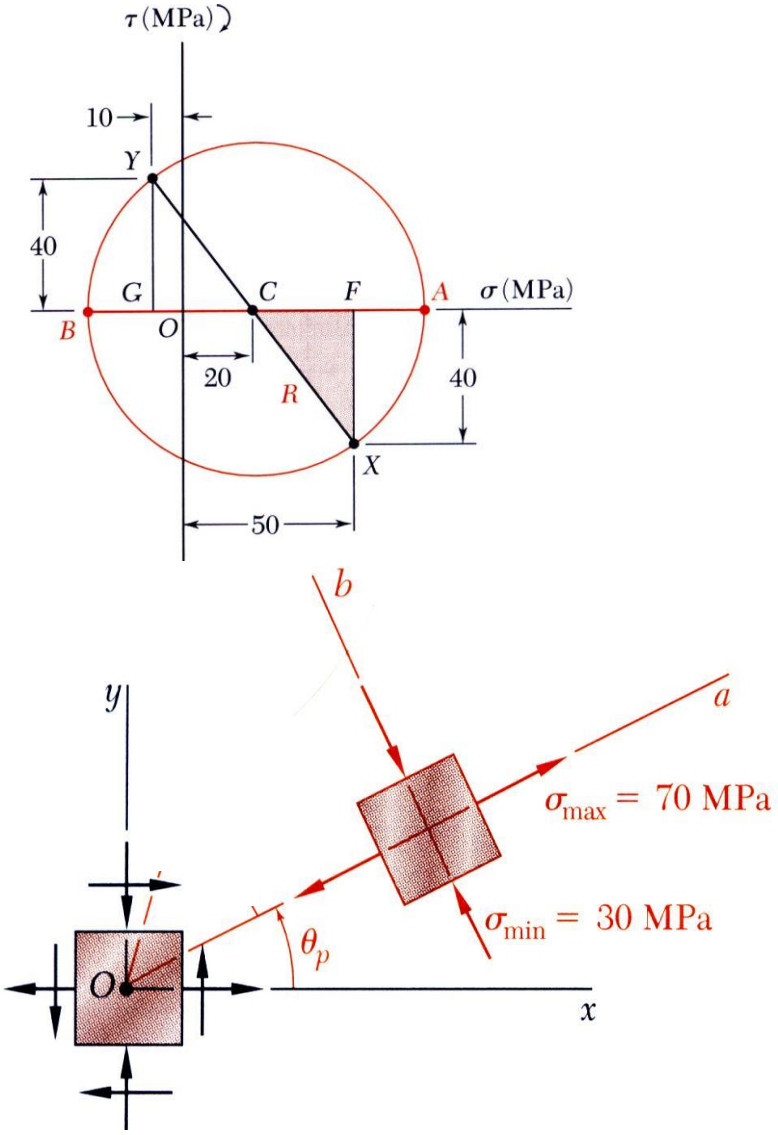
- Construction of Mohr's circle

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{(50) + (-10)}{2} = 20\,\text{MPa}$$

$$CF = 50 - 20 = 30\,\text{MPa} \qquad FX = 40\,\text{MPa}$$

$$R = CX = \sqrt{(30)^2 + (40)^2} = 50\,\text{MPa}$$

Example



- Principal planes and stresses

$$\sigma_{\max} = OA = OC + CA = 20 + 50$$

$$\boxed{\sigma_{\max} = 70\,\text{MPa}}$$

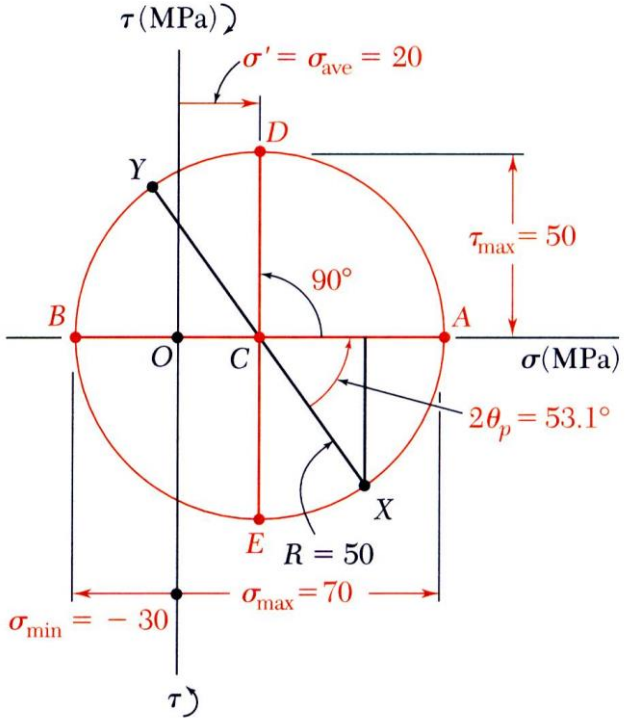
$$\sigma_{\min} = OB = OC - BC = 20 - 50$$

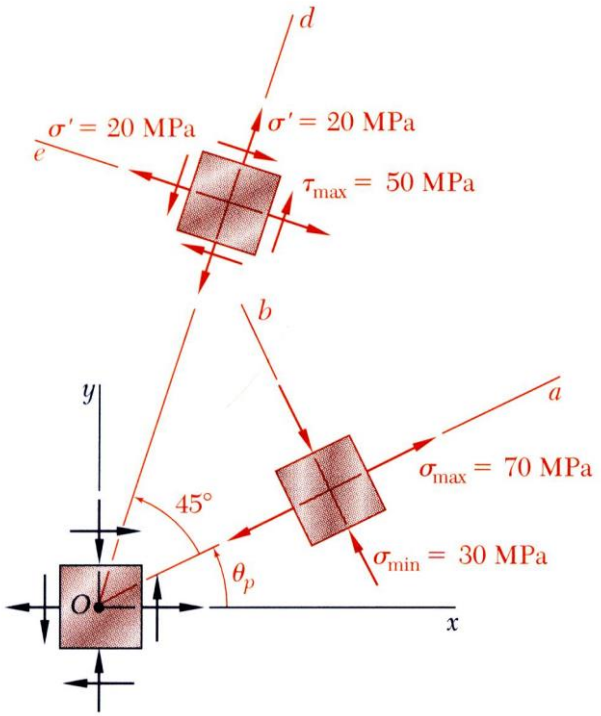
$$\boxed{\sigma_{\min} = -30\,\text{MPa}}$$

$$\tan 2\theta_p = \frac{FX}{CF} = \frac{40}{30}$$

$$2\theta_p = 53.1°$$

$$\boxed{\theta_p = 26.6°}$$





- Maximum shear stress

$$\theta_s = \theta_p + 45°$$

$$\boxed{\theta_s = 71.6°}$$

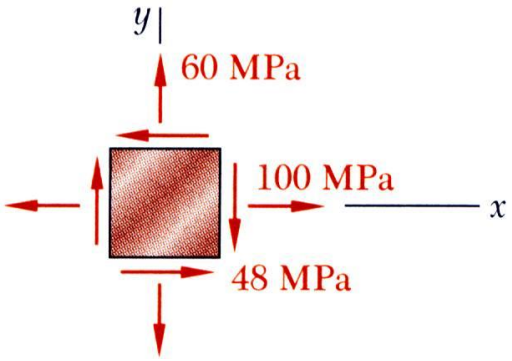
$$\tau_{\max} = R$$

$$\boxed{\tau_{\max} = 50\ \text{MPa}}$$

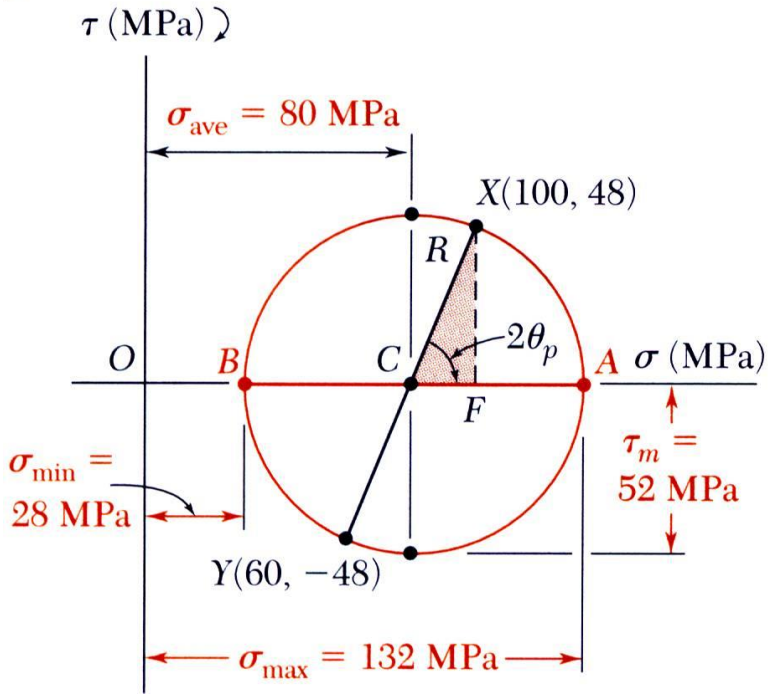
$$\sigma' = \sigma_{ave}$$

$$\boxed{\sigma' = 20\ \text{MPa}}$$

Problem



For the state of stress shown, determine (a) the principal planes and the principal stresses, (b) the stress components exerted on the element obtained by rotating the given element counterclockwise through 30 degrees.

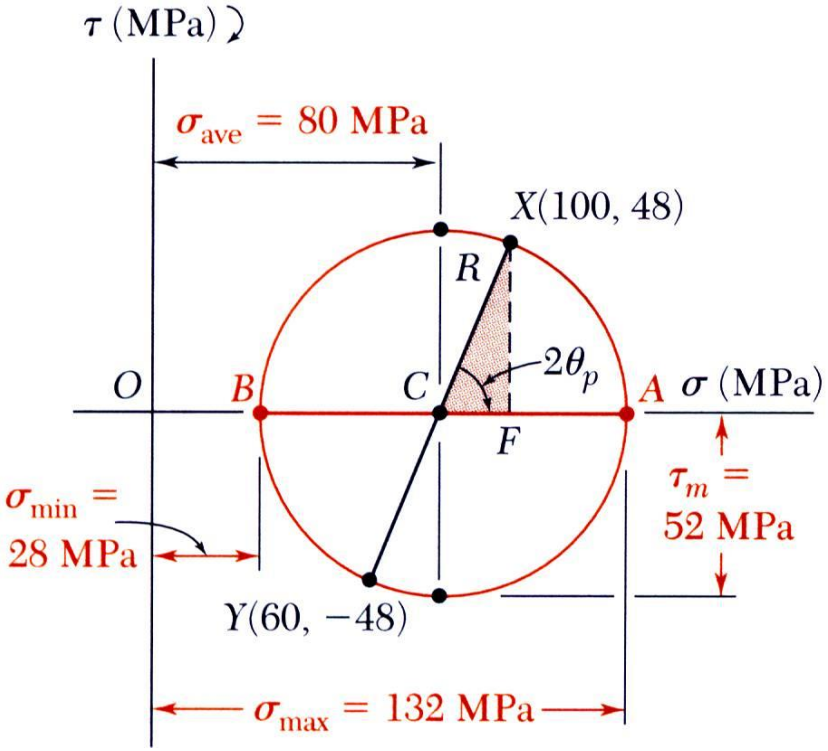


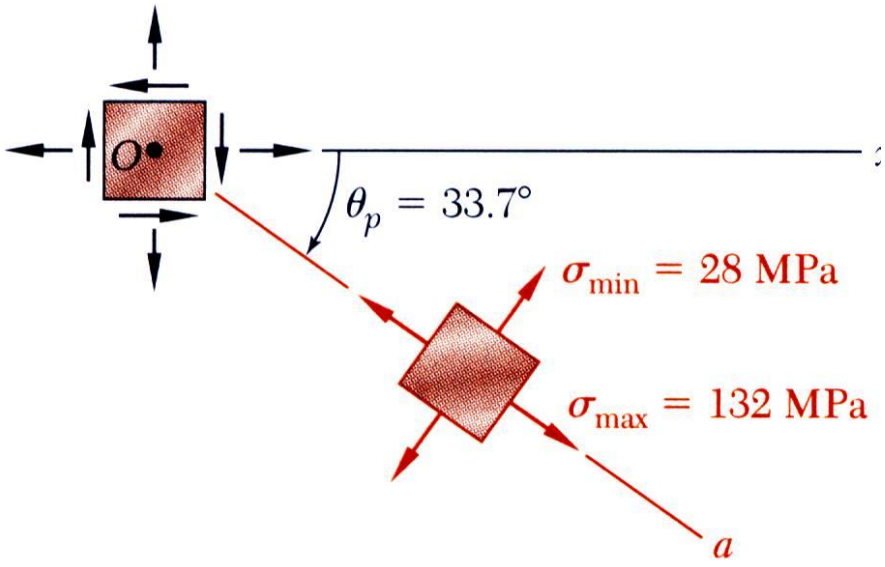
SOLUTION:

- Construct Mohr's circle

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{100 + 60}{2} = 80\,\text{MPa}$$

$$R = \sqrt{(CF)^2 + (FX)^2} = \sqrt{(20)^2 + (48)^2} = 52\,\text{MPa}$$





- Principal planes and stresses

$$\tan 2\theta_p = \frac{XF}{CF} = \frac{48}{20} = 2.4$$

$$2\theta_p = 67.4°$$

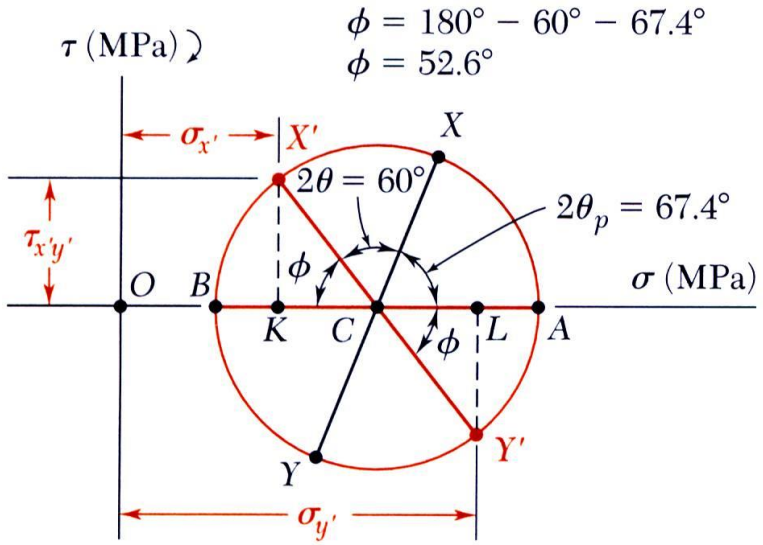
$$\boxed{\theta_p = 33.7° \text{ clockwise}}$$

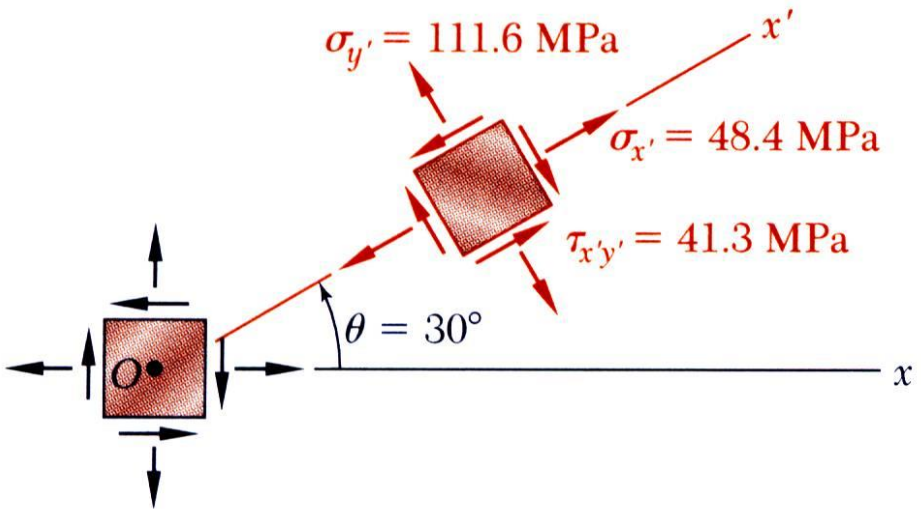
$$\sigma_{max} = OA = OC + CA = 80 + 52$$

$$\boxed{\sigma_{max} = +132\,\text{MPa}}$$

$$\sigma_{max} = OA = OC - BC = 80 - 52$$

$$\boxed{\sigma_{min} = +28\,\text{MPa}}$$





- Stress components after rotation by 30°

  Points X' and Y' on Mohr's circle that correspond to stress components on the rotated element are obtained by rotating XY counterclockwise through $2\theta = 60°$

$$\phi = 180° - 60° - 67.4° = 52.6°$$

$$\sigma_{x'} = OK = OC - KC = 80 - 52\cos 52.6°$$

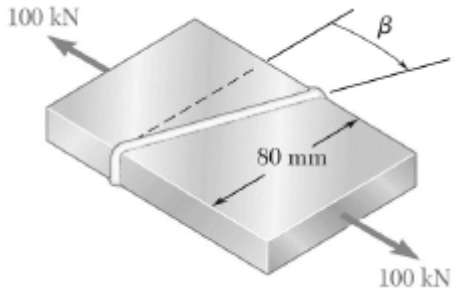
$$\sigma_{y'} = OL = OC + CL = 80 + 52\cos 52.6°$$

$$\tau_{x'y'} = KX' = 52\sin 52.6°$$

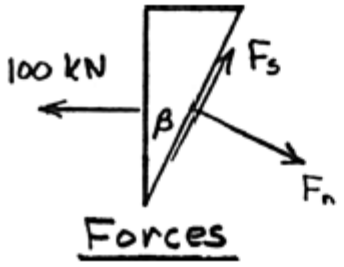
$$\sigma_{x'} = +48.4\,\text{MPa}$$

$$\sigma_{y'} = +111.6\,\text{MPa}$$

$$\tau_{x'y'} = 41.3\,\text{MPa}$$



Two steel plates of uniform cross section 10×80 mm are welded together as shown. Knowing that centric 100-kN forces are applied to the welded plates and that the in-plane shearing stress parallel to the weld is 30 MPa, determine (*a*) the angle β, (*b*) the corresponding normal stress perpendicular to the weld.



Area of weld:

$$A_w = \frac{(10 \times 10^{-3})(80 \times 10^{-3})}{\cos\beta} = \frac{800 \times 10^{-6}}{\cos\beta}\ \text{m}^2$$

(*a*) $\sum F_s = 0$: $F_s - 100\sin\beta = 0$ $\quad F_s = 100\sin\beta\ \text{kN} = 100 \times 10^3 \sin\beta\ \text{N}$

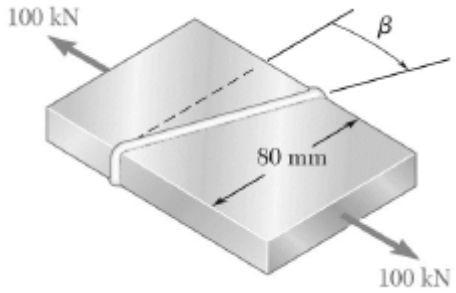
$$\tau_w = \frac{F_s}{A_w} \quad 30 \times 10^6 = \frac{100 \times 10^3 \sin\beta}{800 \times 10^{-6} / \cos\beta} = 125 \times 10^6 \sin\beta\cos\beta$$

$$\sin\beta\cos\beta = \frac{1}{2}\sin 2\beta = \frac{30 \times 10^6}{125 \times 10^6} = 0.240 \qquad \beta = 14.34° \blacktriangleleft$$

(*b*) $\sum F_n = 0$: $F_n - 100\cos\beta = 0$ $\quad F_n = 100\cos 14.34° = 96.88\ \text{kN}$

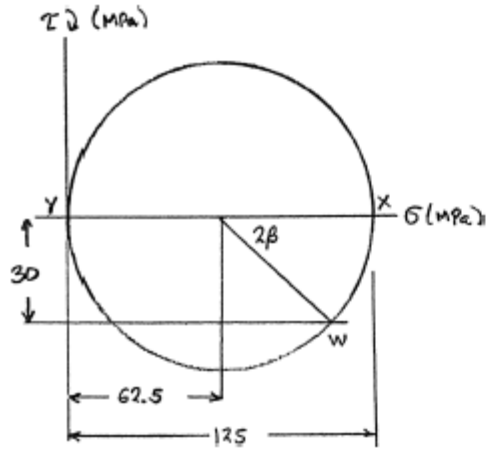
$$A_w = \frac{800 \times 10^{-6}}{\cos 14.34} = 825.74 \times 10^{-6}\ \text{m}^2$$

$$\sigma = \frac{F_n}{A_w} = \frac{96.88 \times 10^3}{825.74 \times 10^{-6}} = 117.3 \times 10^6\ \text{Pa} \qquad \sigma = 117.3\ \text{MPa} \blacktriangleleft$$



Two steel plates of uniform cross section 10×80 mm are welded together as shown. Knowing that centric 100-kN forces are applied to the welded plates and that the in-plane shearing stress parallel to the weld is 30 MPa, determine (*a*) the angle β, (*b*) the corresponding normal stress perpendicular to the weld.

SOLUTION



$$\sigma_x = \frac{P}{A} = \frac{100 \times 10^3}{(10 \times 10^{-3})(80 \times 10^{-3})} = 125 \times 10^6 \text{ Pa} = 125 \text{ MPa}$$

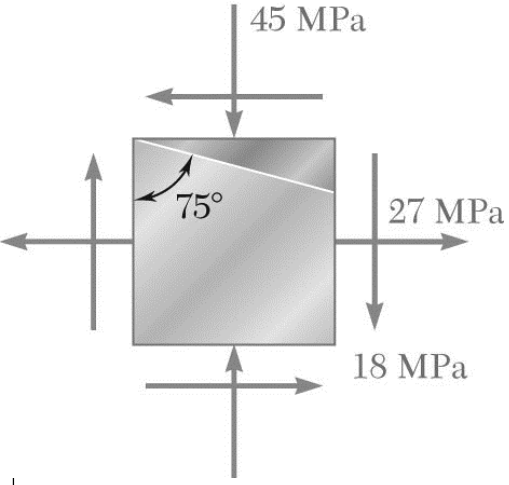
$$\sigma_y = 0 \qquad \tau_{xy} = 0$$

From Mohr's circle:

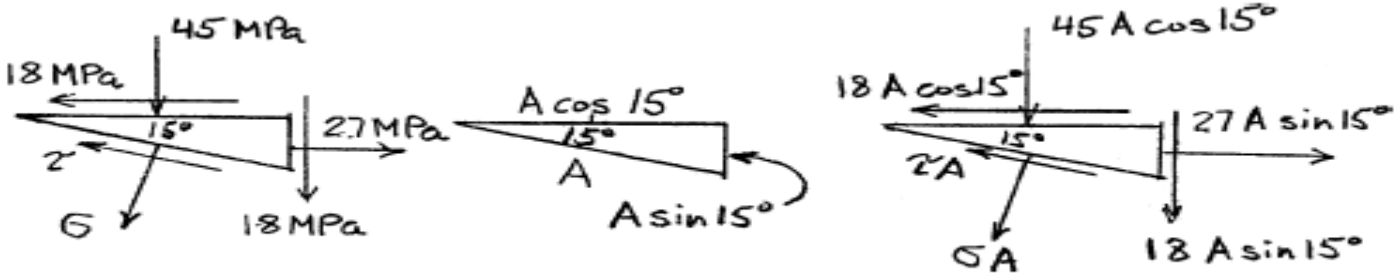
(*a*) $$\sin 2\beta = \frac{30}{62.5} = 0.48 \qquad \beta = 14.3° \blacktriangleleft$$

(*b*) $$\sigma = 62.5 + 62.5 \cos 2\beta$$

$$\sigma = 117.3 \text{ MPa} \blacktriangleleft$$



For the given state of stress, determine the normal and shearing stresses exerted on the oblique face of the shaded triangular element shown. Use a method of analysis based on the equilibrium of that element.



Stresses Areas Forces

$\sigma = -49.2$ MPa $\tau = 2.41$ MPa